OCL PRAVILA

Zadatak radile:

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1. Lijeva i desna strana iskaza, a i izraza moraju da budu istog tipa - kompatibilnost (mogu da se mijesaju float i integer)

```
context Choice:
inv:
        (self.left.expressionPart -> select(te | te.oclIsTypeOf(TypedElement)) -> exists(te | te.dataType.oclIsTypeOf(Boolean))
        implies self.right.expressionPart -> select (te | te.oclIsTypeOf(TypedElement)) -> forAll(te | te.dataType.oclIsTypeOf(Boolean)))

context AssignmentStatement:
inv:
        (self.variable.dataType.oclIsTypeOf(Boolean) implies
        self.expression.expressionPart -> select(ep | ep.oclIsTypeOf(TypedElement)) -> forAll (te | te.dataType.oclIsTypeOf(Boolean))
```

2. Da se varijabla ne može koristiti prije deklaracije

```
context Expression
inv NoUseBeforeAssignment:
        self.expressionPart -> select(ep | ep.oclIsTypeOf(Variable)) -> forAll(v | AssignmentStatement::allInstances() -> select(as | as.variable = v) -> notEmpty())
```

3. Sve varijable moraju da imaju unique imena

```
context Variable
inv: Variable.allInstances()->forAll(v1, v2 | v1 <> v2 implies v1.name <> v2.name)
```

4. Broj operanada mora da bude jednak broju TypedElemenata - 1

```
context Expression
inv:
  (self.expressionPart->select(ep | ep.oclIsTypeOf(ArithmeticOperand) or ep.oclIsTypeOf(ComparationOperand))->size()) + 1 =
  self.expressionPart->select(ep | ep.oclIsTypeOf(TypedElement))->size()
```

5. Inicijalne vrijednosti za različite tipove varijabli prilikom deklaracije

```
context Variable::value : Integer
init: if self.dataType.ocllsTypeOf(Integer) then self.value = 0 endif

context Variable::value : Real
init: if self.dataType.ocllsTypeOf(Float) then self.value = 0.0 endif

context Variable::value : Boolean
init: if self.dataType.ocllsTypeOf(Boolean) then self.value = false endif
```

6. Aritmetičke operacije se primenjuju samo na Float ili Integer tipove

```
context Expression
inv:
  self.expressionPart->exists(oclIsTypeOf(ArithmeticOperand)) implies
  not self.expressionPart->select(e | e.oclIsTypeOf(TypedElement))->exists(te |
te.dataType.oclIsTypeOf(Boolean))
```

7. Ako je Statement End onda nema next atribut

```
context Statement
inv ProperSequence: self.next->notEmpty() or self.oclIsTypeOf(End)
```

8. Nad bool tipovima se moze raditi samo == i != komparacija

```
context Choice
inv:
   self.left.expressionPart->select(e | e.oclIsTypeOf(TypedElement))->exists(te |
te.dataType.oclIsTypeOf(Boolean))
   implies
      (self.comparationOperand.oclIsTypeOf(Equal) or
self.comparationOperand.oclIsTypeOf(NotEqual))
```

9. U Choice ni left ni right ne smije da ima ComparationOperand (da li levi i desni expresion ima comparation)

```
context Choice
inv:
   self.left.expressionPart->forAll(e | not e.oclIsTypeOf(ComparationOperand)) and
   self.right.expressionPart->forAll(e | not e.oclIsTypeOf(ComparationOperand))
```

10. Expression treba da ima najmanje 1 Operand i 2 TypedElement-a

```
context Expression
```

```
inv:
        self.expressionPart -> select(oclIsTypeOf(ArithemeticOperand) or
oclIsTypeOf(ComparationOperand)) -> size() > 0

context Expression
inv:
        self.expressionPart -> select(oclIsTypeOf(Variable) or oclIsTypeOf(Constant)) ->
size() > 1
```

11. Choice nextIfTrue i nextIfFalse ne smiju biti empty

```
context Choice
inv: not self. nextIfTrue.oclIsUndefined() and not  self.nextIfFalse.oclIsUndefined()
```

12. Moze da postoji samo jedan Start i End statement

```
context Start
inv: Start::allInstance()->size()=1

context End
inv: End::allInstance()->size()=1
```

13. Svaki Variable element mora da ima DataType (ne moramo da dodamo za value pošto smo napisale pravilo za inicijalne vrijednosti)

```
context Variable
inv: not self.dataType.oclIsUndefined()
```

14. Svaki Constant element mora da ima DataType i value

```
context Constant:
inv: not self.dataType.oclIsUndefined() and not self.value.oclIsUndefined()
```

15. Svaki Comment mora da ima neprazan text

```
context Comment
inv: not self.text.oclIsUndefined() and self.text.size() > 0
```

16. Atribut next kod Statement-a ne moze da pokazuje sam na sebe (na isti Statement)

```
context Statement
inv: self.next <> self
```